



ENT-164 · Intro to Making

Tufts
UNIVERSITY

Derby Entrepreneurship
Center at Tufts

CLASS 5 · WEEK 5 · FALL 2026

Electronics

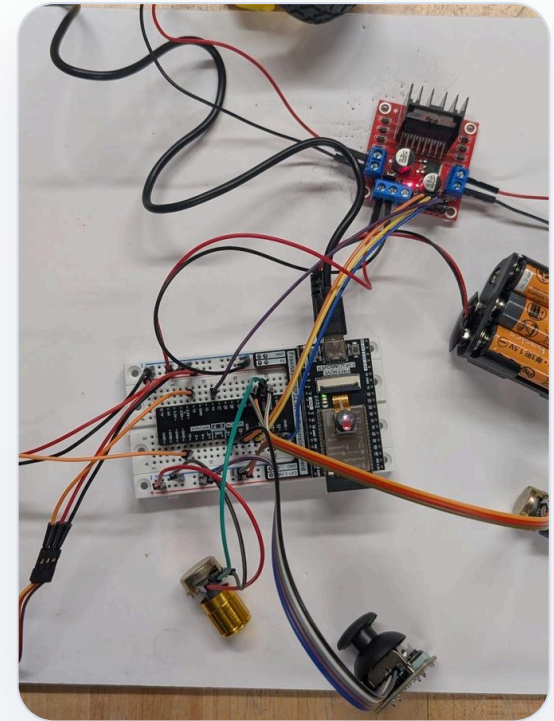
Today your robot gets a brain. We wire motors, sensors, and controls on the breadboard, install MicroPython, and make AI our coding partner.

Thu, Oct 8

JCC 301 / Nolop

Robo Challenge

Bring your ESP32 kit



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Tufts University · Gordon Institute

Seven weeks from sketch to race day

Today is Week 5. Basic Electronics and Microcontrollers. Here's how the robot build comes together.

Week 2

Ideation and Prototyping

Sketch concept variations; build a foam core mockup.

Week 3

From Physical to Digital: 2D

Create a laser-ready 2D design; cut structural parts.

Week 4

From Physical to Digital: 3D

Model 3D parts; 3D print functional parts.

Week 5

Basic Electronics and Microcontrollers

Wire motors, sensors, and controls; program with MicroPython + AI.

Week 6

AI Coding Agents

Connect to a wireless network; build a remote control interface.

Week 7

Final Robot Assembly and Testing

Final integration and test runs.

Week 8

Robo Challenge and Final Project Kickoff

Demonstrate the robot with mechanical systems and motors mounted.

TODAY

Week 5 · Electronics. Wire motors, sensors, and controls on a breadboard, and program basic movements using MicroPython and AI.

Today the robot gets a brain

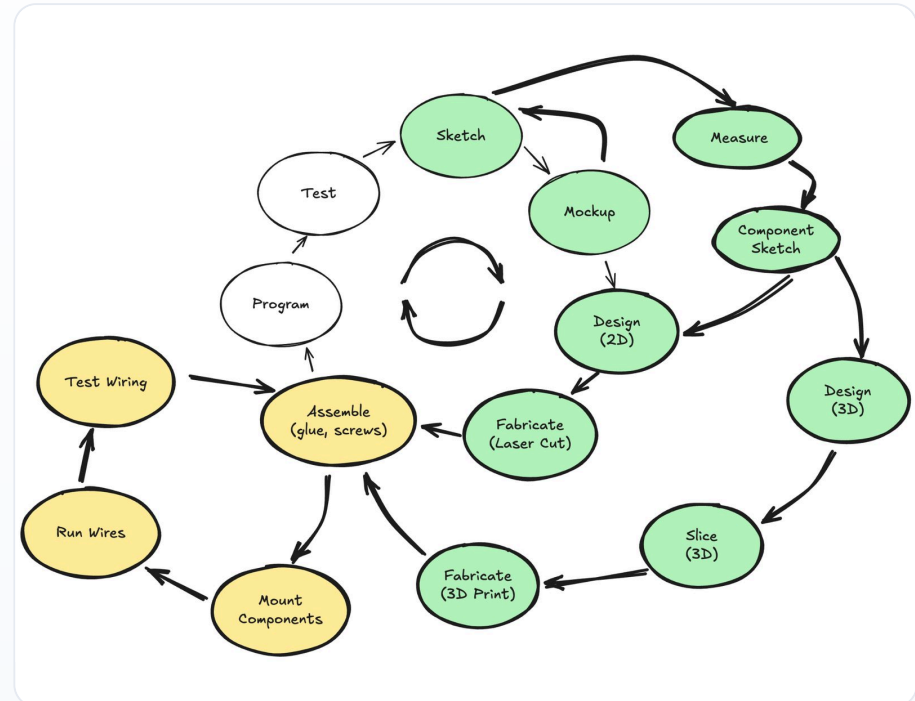
The mechanics are nearly done. Today we bring the electronics to life.

TODAY'S PLAN

- Meet your kit and the ESP32 board
- Install the MicroPython operating system
- Blink an LED with opencode
- Move a servo with AI-generated code
- Wire the robot at Nolop

WHERE WE ARE IN THE BUILD

Sketch, mockup, design, and fabrication are behind you. Now we assemble, run wires, test the wiring, program, and test again.



What's in your **personal kit**?

Freenove FNK0046 Super Starter Kit for ESP32 WROVER. Every part the electronics project needs.

The brain

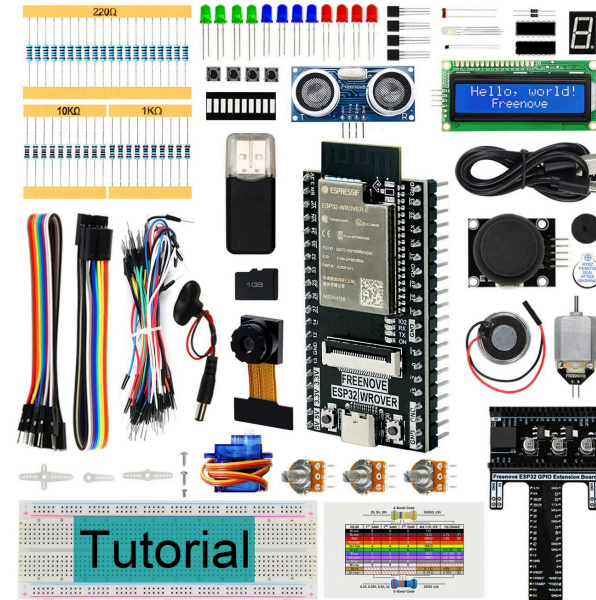
ESP32-WROVER board, GPIO extension board, and a solderless breadboard.

The basics

Jumper wires, LEDs, resistors (220 Ω , 1K, 10K), push buttons, USB cable.

The fun stuff

LCD and 7-segment displays, buzzers, joystick, potentiometers, servo, TT motor + wheel, motor driver, ultrasonic sensor, camera.



NOTE

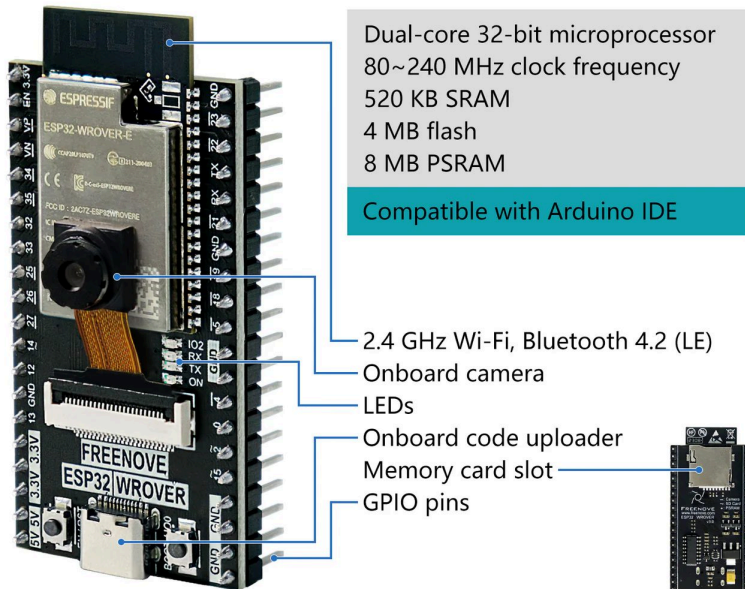
No paper tutorial. The download link for the tutorial and code is **on the box**.

Meet the ESP32-WROVER

A powerful, small controller with onboard wireless and camera.

ESP32-WROVER Board

A powerful, generic and small-sized controller with onboard wireless and camera.



SPECS

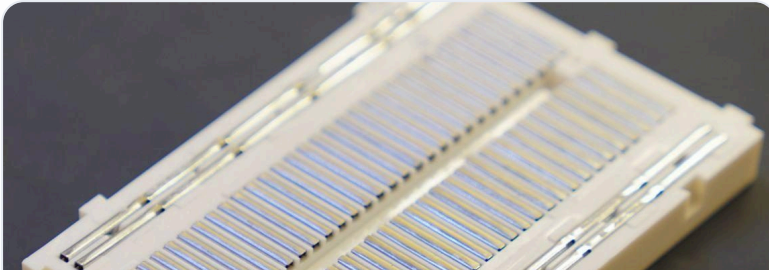
- Dual-core 32-bit microprocessor, 80–240 MHz
- 520 KB SRAM · 8 MB PSRAM · 4 MB flash
- 2.4 GHz Wi-Fi and Bluetooth 4.2 LE
- Onboard camera, LEDs, and code uploader
- Memory card slot and GPIO pins

WHAT IT CONTROLS

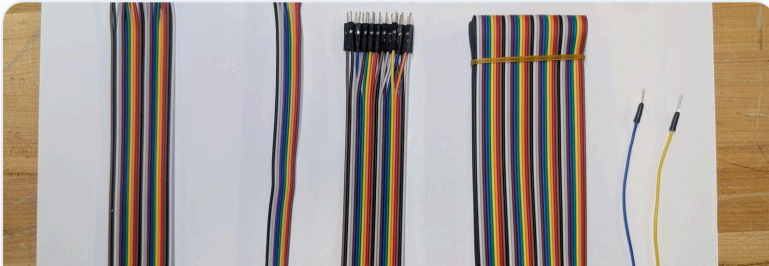
Motors through the driver, servos, joystick, potentiometers, sensors, and displays. The brain of your robot.

How the **breadboard** works

No soldering. Push parts and wires into the holes and they connect.



A solderless breadboard. Rows of connected holes



Jumper wires. Male–male, male–female, female–female

Rows

Each row of five holes is connected horizontally, and the center channel separates the two sides.

Power rails

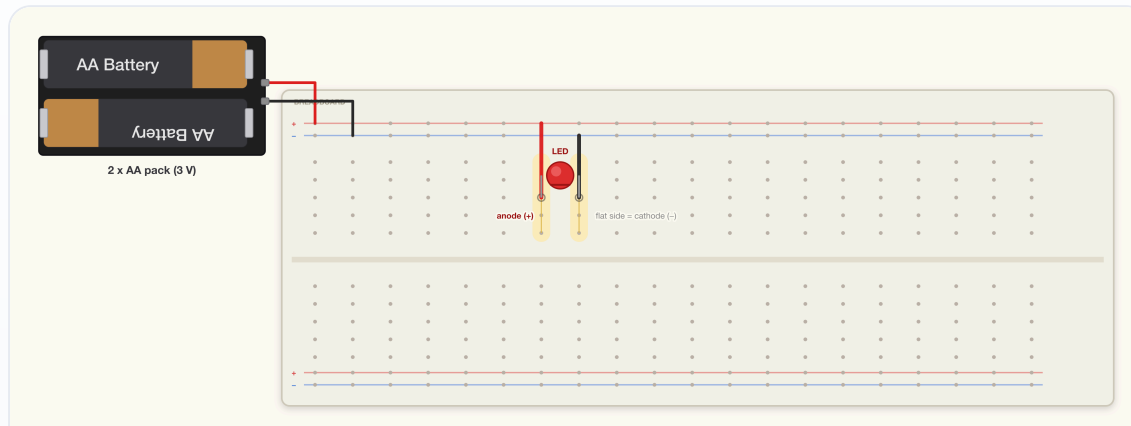
The vertical rails along the edges carry 5V, 3.3V, and GND to every row.

Pin rows

Your ESP32 plugs into the GPIO extension board: any of the horizontal holes in a pin's row connects to that pin. For example, pin 34.

An LED with no resistor

The LED straight across the rails. It lights, it is very bright, and it is being overdriven.



Follow the steps

The diagram lists them, and the class page has the same one at full size.

Wire it

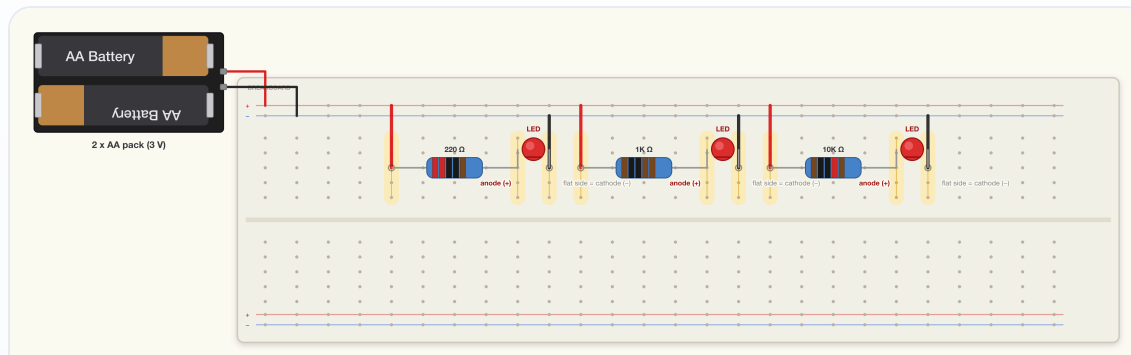
- Pack at the left: red lead to the + rail, black to the -
- LED: long leg in column 6, short leg in column 7, row c
- Red jumper: the + rail to column 6. Black jumper: column 7 to the - rail

On for a second

Connect it, look, then pull the red lead back out. Then take it apart and build exercise 2.

220 ohm, 1 kohm, 10 kohm

The same LED with three different resistors. Bright, dimmer, barely on.



Follow the steps

Three branches in one build, so you can compare them side by side.

Wire it

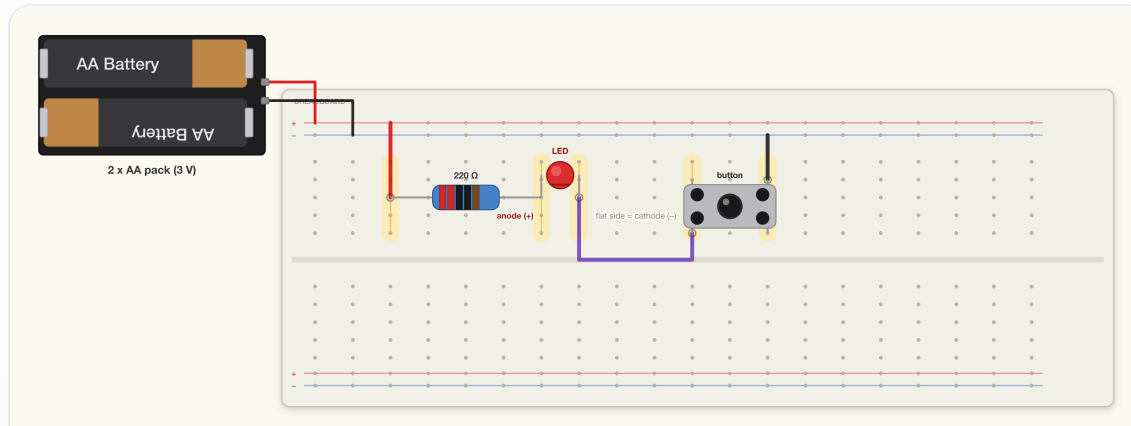
- A resistor and LED per branch: 220 Ω , then 1K Ω , then 10K Ω , row c
- Red jumpers: the + rail to each resistor's left leg
- Black jumpers: each LED back to the - rail

More resistance, less current

The resistor decides how much current the LED gets, and the current decides how bright it is.

Add a push button

The 220 Ω circuit with a switch in the path. Press to light the LED, let go and it goes dark.



Follow the steps

The button goes in the path, not beside it.

Wire it

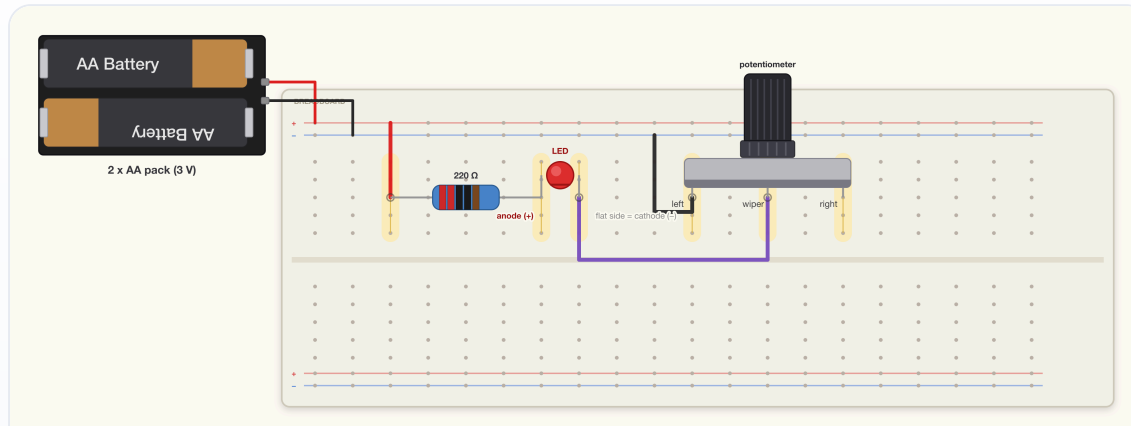
- Resistor: the + rail to column 2, body to column 6
- LED: columns 6 and 7, row c
- Button: columns 10 and 12, rows b and e
- Purple jumper: column 7 to column 10. Black: column 12 to the - rail

Two states, nothing between

Pressed or not. On or off. That is a digital signal, and a pin on the board reads it the same way.

Swap in a potentiometer

Take the button out and put the knob in its place. Turning it makes the LED brighter and dimmer.



Follow the steps

One end of the track takes the current, the wiper passes it on.

Swap it

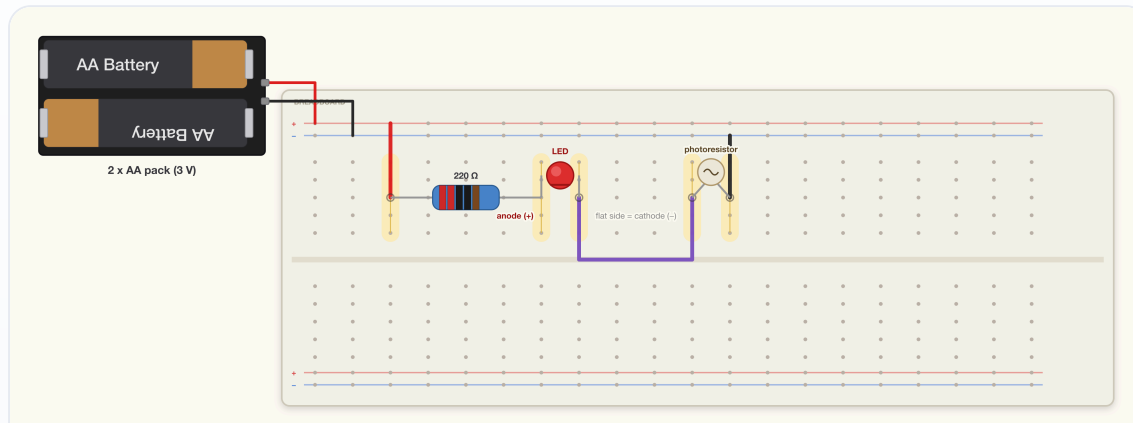
- Take the button out
- Pot: legs in columns 10, 12 and 14, row c
- Purple jumper: column 7 (the LED) to column 12, the middle leg
- Black jumper: column 10, the end of the track, to the - rail

A range, not two states

The wiper taps the track wherever you turn it, so the resistance changes smoothly. That is analogue.

Swap in a photoresistor

Take the knob out and put a light sensor in its place. The LED follows the light.



Follow the steps

The sensor takes the knob's place in the same path.

Swap it

- Take the pot out
- Photoresistor: its two legs in columns 10 and 11, row c
- Purple jumper: column 7 to column 10
- Black jumper: column 11 to the - rail

Light is the knob

More light, less resistance, more current, brighter LED. Cover it and the LED fades.

Signals **travel on the wire**

A voltage, held or changed, read at the other end. That is the whole conversation between a microcontroller and a peripheral.

01

Digital

Two states: 3.3 V or 0 V. Your button. A pin reads it, and writes it.

02

Analogue

A voltage anywhere in the range. Your knob and your light sensor. An ADC pin turns it into a number.

03

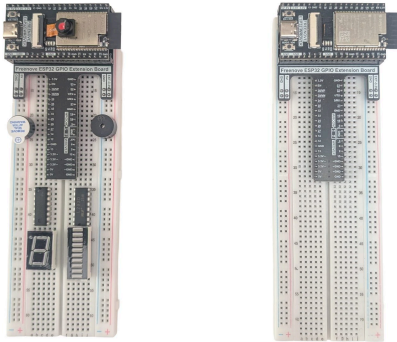
Patterns

Switch a pin fast, and the average is what matters: that is how brightness and servo angles are sent. Modules that pass numbers do the same, in a pattern both ends agree on.

Next: the ESP32 does this in code. The same LED, the same button, the same knob, with a pin doing the reading and the writing.

Prep your board

Three steps before the first circuit.



1 · Remove the pre-installed demo parts

02

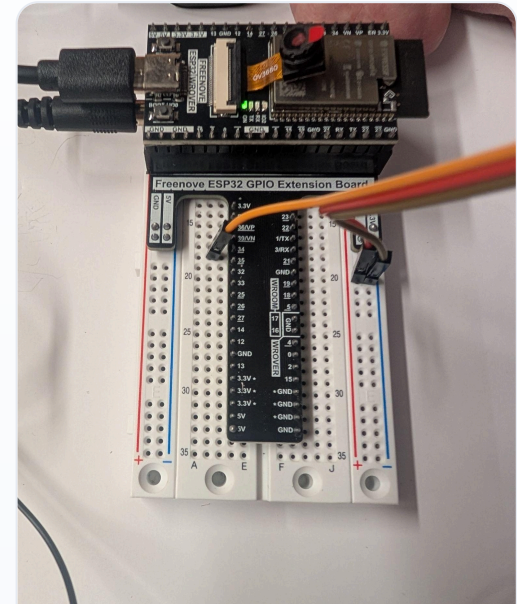
Seat the board

Clip the ESP32 and its GPIO extension board across the breadboard's center channel.

03

Read the labels

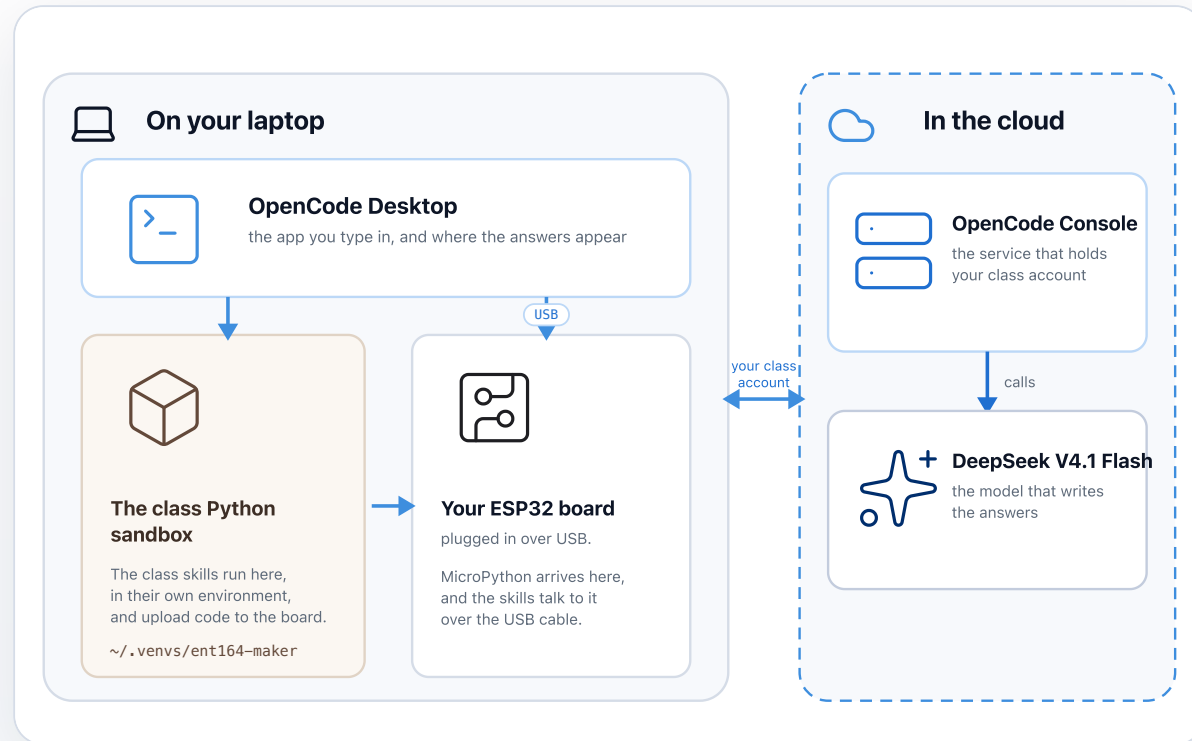
5V, 3V3, GND, and the numbered GPIO pins are printed on the extension board. Check before you plug anything in.



Pin labels on the extension board

Where each part runs

Your laptop and the class cloud — what runs where. The setup guide connects them.



Everything on the left runs on your laptop; the class account and the model live in the cloud.

Install opencode with the **setup guide**

Join the class workspace, install the app, add the class skills. One guide, macOS and Windows.

The setup guide

`tuftsmaker.github.io/ENT-164/opencode-setup`

It walks through

- Accept the class invitation, then sign in
- Install the app and choose the class model
- Add the class skills, then ask it to flash the board



The MicroPython snake needs a plane. Your ESP32 board.

How it works

You asked opencode to blink an LED. Here's what happened behind the scenes.

BLINK.PY

```
from machine import Pin
import time

led = Pin(2, Pin.OUT)

while True:
    led.on()      # Turn LED on
    time.sleep(1) # Wait 1 second
    led.off()     # Turn LED off
    time.sleep(1) # Wait 1 second
```

WHAT THE AGENT DID

- Wrote MicroPython for the onboard LED on GPIO 2
- Put the MicroPython runtime on the board, then copied the program in over USB
- Started it — the LED blinks the moment the program lands

LOOK INSIDE

Ask opencode: “Show me the code you just wrote and explain it line by line.”

Then push further: what is a Pin? What does sleep() do? Change the timings.

CHALLENGE

Can you make it blink faster or slower?

Wire the external LED

Ask opencode to show how to blink an external LED — it draws the whole circuit, wires included.

CLICK A WIRE

Every wire is clickable: click one to see where it connects, end to end. The numbered build order and the notes come with the diagram.

THEN THIS

Ask opencode: “Generate a wiring table that summarizes the connections.”

A wiring table makes it easy to check your board before you power it on.

Add an external LED — Freenove ESP32-WROVER (FNK0046)
 Four steps, two jumper wires — the ESP32 board stays off the breadboard

Legend:
 ■ signal — GPIO 2 ■ ground — GND ● board pin ■ one connected net (5 holes) ● build order

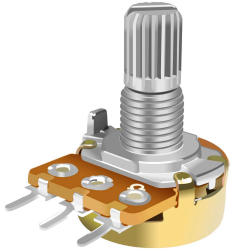
Steps

1. Red jumper: board GPIO 2 → breadboard column 2, top row.
2. Resistor 220 Ω: one leg in column 2, other leg in column 6 (same row).
3. LED: long leg (anode) in column 6, short leg (cathode) in column 7, same row.
4. Black jumper: breadboard column 7, bottom row → board GND (right header, pin 17).

Never wire an LED straight to 3.3 V or a GPIO without the resistor — it will burn out.
 The five holes in one breadboard column are already connected — that is the whole point of a breadboard.
 GPIO 2 also drives the onboard LED, so both light together. Move the red wire to GPIO 4 for a separate LED.

Knobs, sticks, and **sensors**

Your robot reads the world through inputs. Here's how two common ones wire up.



Potentiometer. A knob the ESP32 can read

POTENTIOMETER

- Left pin → GND
- Middle pin (wiper) → ESP32
- Right pin → 3V3



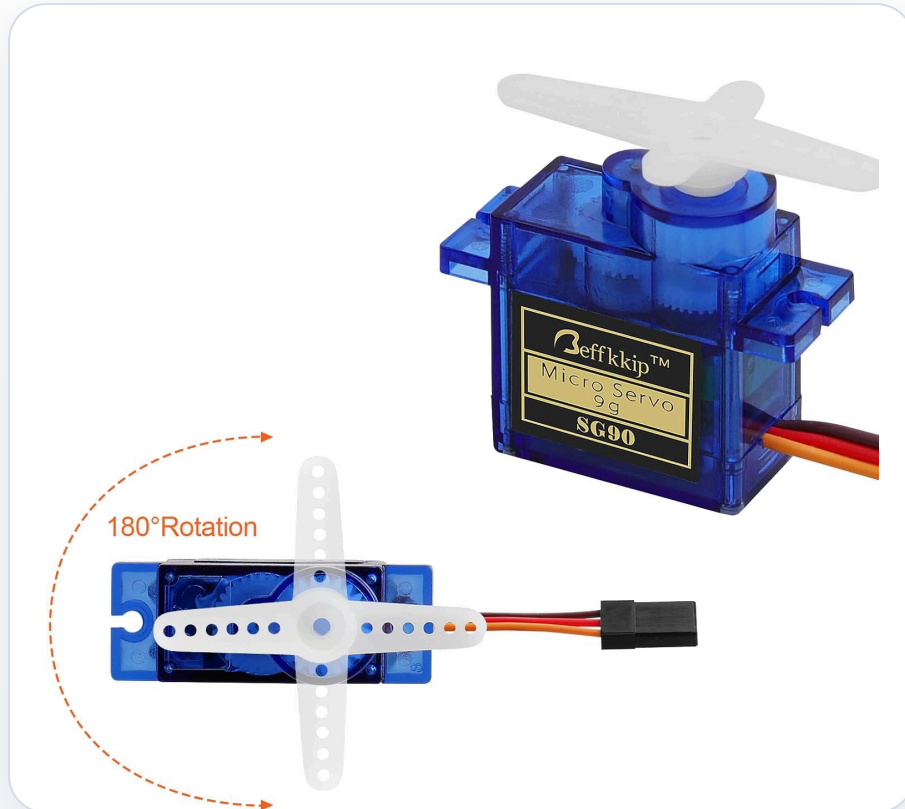
KY-023 joystick. Two axes plus a button

KY-023 JOYSTICK

- GND → GND
- +5V → 3V3
- VRx (X-axis) → ESP32
- VRy (Y-axis) → ESP32
- SW (button) → ESP32

Move an arm with a servo

The SG90 micro servo rotates about 180°. Perfect for arms, gates, and grippers.



SERVO WIRING

- Brown / Black → GND
- Red → 5V
- Orange / Yellow → ESP32

ON THE ROBOT

Servo 1 and Servo 2 wire up the same way. Match the colors, not the memory.

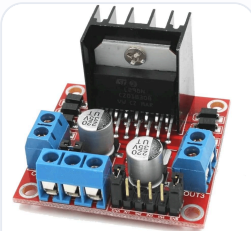
Spin the wheels with the L298N

The driver is the muscle: the ESP32 sends signals, the L298N sends power to the motors.



TT Motor

DC gearbox
motor + wheel



L298N driver

Dual H-bridge
motor driver

L298N WIRING

- 12V → battery pack
- 5V → logic power
- GND → common ground
- IN1, IN2 → ESP32 (left motor)
- IN3, IN4 → ESP32 (right motor)
- OUT1, OUT2 → left motor (red, black)
- OUT3, OUT4 → right motor (red, black)

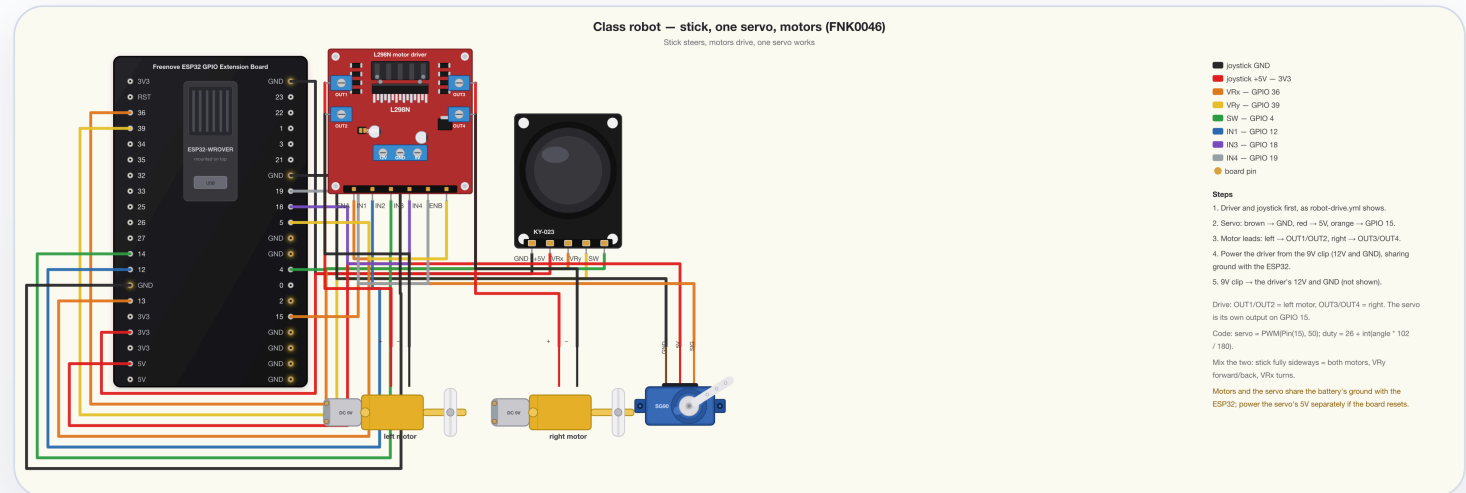
Robot wiring example

Stick, one servo, two drive motors, one driver, one brain.

READ IT LIKE A MAP

- Power comes from the battery pack
- Motors run through the L298N driver
- Servo and controls go to GPIO pins

Wire from a diagram, never from memory, and ask AI to explain any wire you don't recognize.



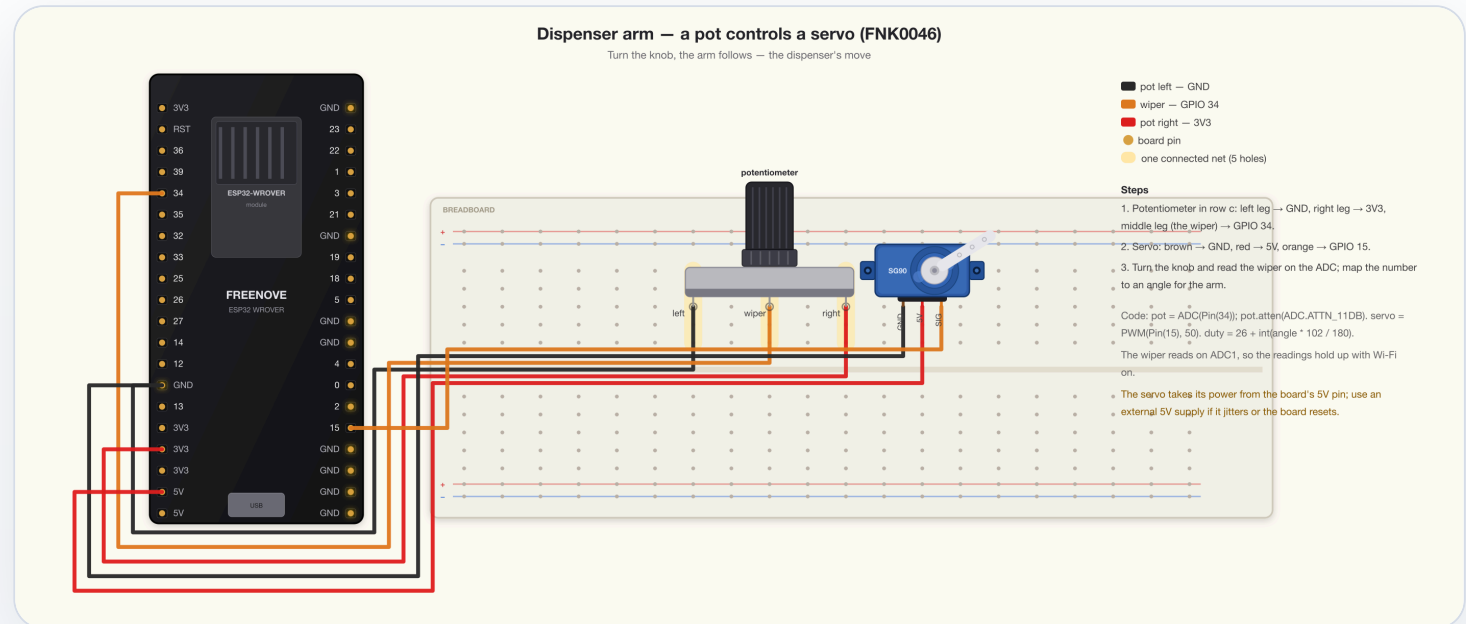
Dispenser wiring example

A pot and a servo — the smallest complete circuit: one input, one output.

READ IT LIKE A MAP

- The pot's wiper reads on GPIO 34 — an analogue input
- The servo takes its signal on GPIO 15
- Turn the knob and the arm follows

Build the dispenser's move on its own first, then mount it on the robot.



Wire your robot **one part at a time**

Mechanical assembly review, then AI-assisted wiring at the makerspace.

TODAY AT NOLOP

- Mechanical assembly review
- Use AI to wire robot components one by one
- Mount parts and run wires after functional testing

TIPS

- Do not generate all the code and wiring in one prompt.
- Add components step by step. Test and verify with each iteration.
- Provide detailed model numbers for the parts.

NAME YOUR PARTS EXACTLY

- DC Gearbox Motor. "TT Motor"
- BOJACK L298N Motor DC Dual H-Bridge Motor Driver
- KY-023 Control Stick
- SG90 9g Micro Servo Motor
- Freenove ESP32 WROOM

AI-assisted electronics project

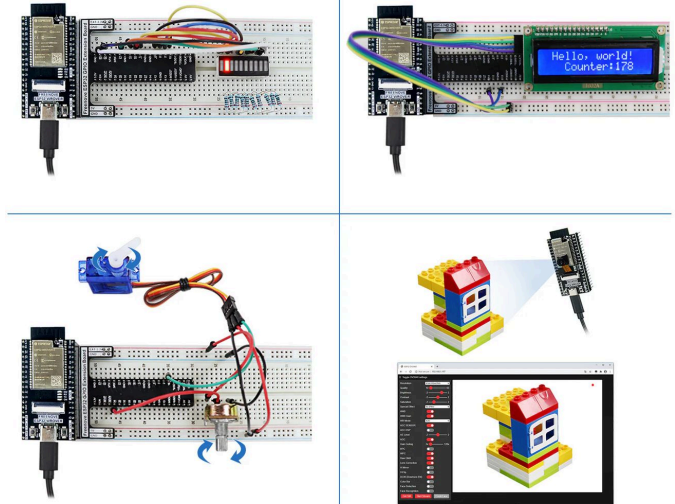
Design and build a fun electronics project using your ESP32 kit. With AI as your coding partner.

REQUIREMENTS

- At least 1 input (sensor, button, etc.) + 1 output (LED, display, buzzer, etc.)
- AI-generated MicroPython code, tested and refined

THE PROCESS. ALL IN ONE CHAT

- **Brainstorm:** attach the parts list image, ask AI for project ideas
- **Build:** get wiring instructions and code from AI
- **Test & iterate:** describe problems, share errors, refine with AI's help



Need ideas? The kit tutorial has example projects and code to borrow from.

What you'll hand in

01

Your AI conversation

Upload a copy of your chat — opencode conversations can't be linked. One continuous conversation, start to finish.

02

Video demo

A demo of the working project with a voice-over explanation.

DEMO Bring your electronics project to class for a demo.

GRADING You're graded on **collaboration and learning**, not coding skills. Keep one chat, iterate often, and ask lots of questions.

Physical Making **reflection**

A short written reflection on the making so far — foam core mockups, laser cutting, 3D printing.

01

Catch the surprise

Find a moment across the three builds when the material didn't do what you pictured. What did you notice, and what did you do next?

02

Think on your feet

Point to a decision you made mid-build without stopping to plan it. What were you reacting to?

03

Look back

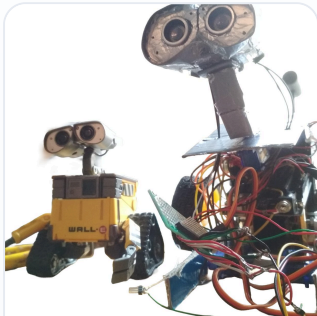
Across the three projects: what would you keep exactly as it is, and what would you change — and why?

HOW TO WRITE IT

Ground every answer in a specific moment from your builds. A few honest paragraphs beat a polished essay.

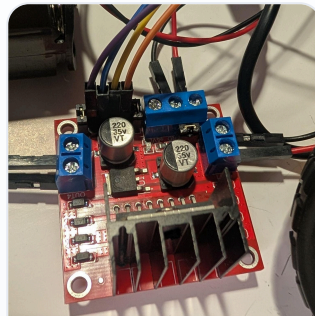
Robots under construction

Wiring, testing, and mounting. The messy middle of the build.



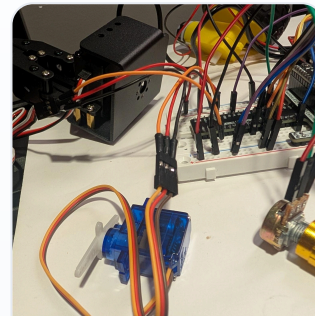
Wire by wire

The robot takes shape.



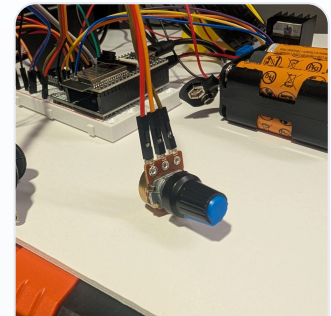
Motor driver

Powered and wired in place.



Test rig

Test each part before mounting.



Controls

Knobs wired and verified.

BEFORE YOU GO

Questions?

Next week: AI Coding Agents. Connect your microcontroller to a wireless network and build a remote control interface.

WHERE TO FIND US

Office hours

go.tufts.edu/meet_tvdv

Email

thomas.van_de_velde@tufts.edu

Course materials

tuftsmaker.github.io/ENT-164